

Name: _____

Class: _____



Mr. Rawson • GCSE Computer Science

Year 11 2026

Programming Challenge: Variables and Constants

Using Constants for a Tax Calculation (HARD) — Answer Key

https://isaacomputerscience.org/questions/varconst_hard

- Write a Python program that calculates the total price of an item using a constant tax rate.
- The price before tax is stored in a variable, and a constant value is used for the tax rate.
- The program should calculate the total price by adding the tax to the original price.

You need to:

1. Declare and initialise a variable for the price of the item.
2. Use a constant for the tax rate (e.g. 20%).
3. Calculate the total price including tax.
4. Display the result.

Example Output:

The total price including tax is 60.

Previous Knowledge Needed

- Constants, named in CAPITALS (line 4)
- Variables (line 8)
- Multiplying with * and adding with + (line 12)
- Printing mixed values with commas (line 21)

Alfred's answer, v2.

```
1 # 2. Use a constant for the tax rate = TAX_RATE =
2 # a constant never changes, so its name is in CAPITALS
3 # (e.g. 20%) = 20 / 100 = 0.2
4 TAX_RATE = 0.2
5
6 # 1. Declare and initialise a variable = price_before_tax =
7 # for the price of the item (before tax) = 50
8 price_before_tax = 50
9
10 # adding the tax = first work out the tax: price_before_tax * TAX_RATE
11 # 50 x 0.2 = 10.0
12 tax = price_before_tax * TAX_RATE
13
14 # 3. Calculate the total price including tax = price_before_tax + tax
15 # 50 + 10.0 = 60.0
16 total_price = price_before_tax + tax
17
18 # 4. Display the result = print()
19 # 0.2 is a decimal, so the total shows as 60.0
20 # The total price including tax is 60.0 = print(..., total_price)
21 print("The total price including tax is", total_price)
```